

EXPERIMENT 30

Threat-Indicator Network Log Analyzer

AIM

To compare network source IP addresses with predefined threat indicators and summarize matching connection records.

ALGORITHM

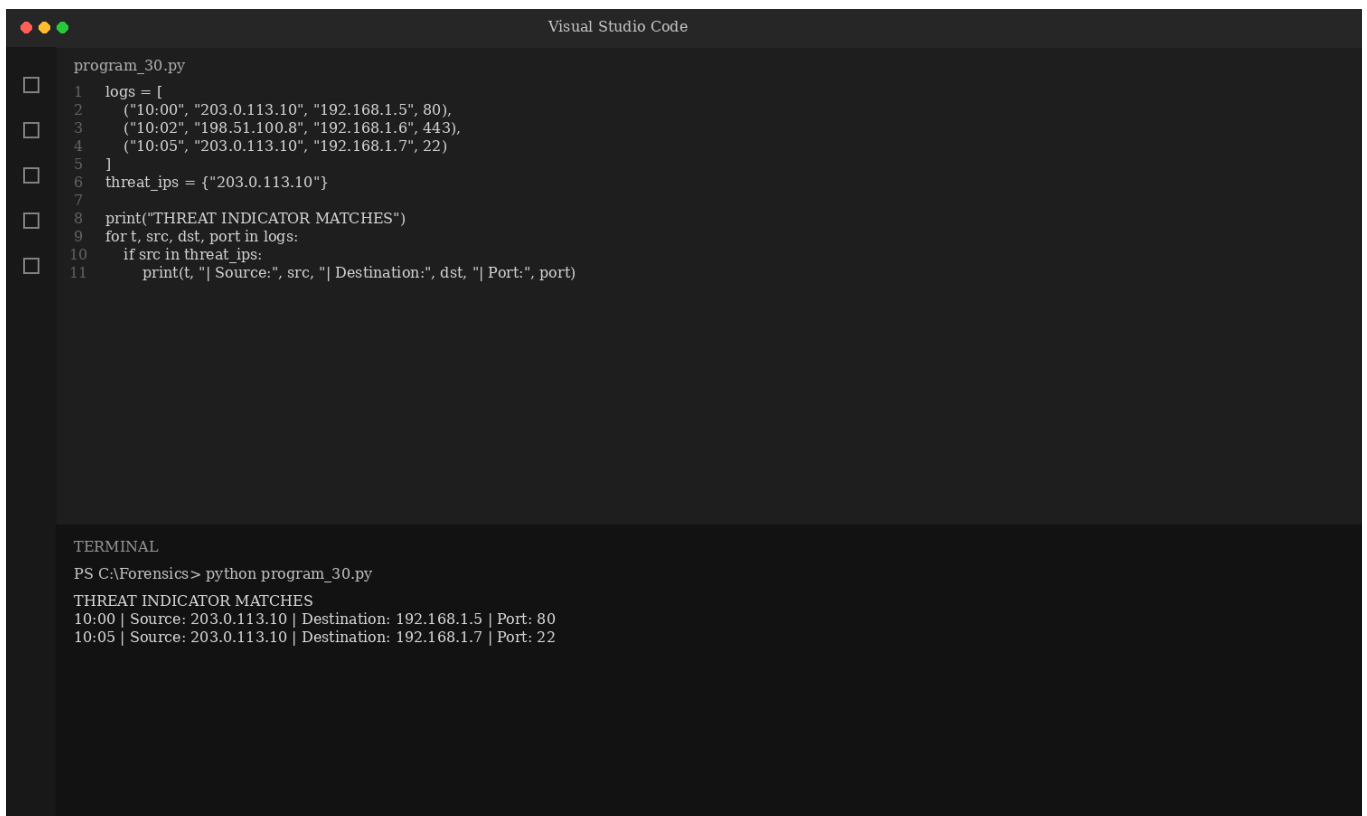
1. Create simulated network records.
2. Create a list of known threat IP addresses.
3. Compare each source IP with the threat list.
4. Collect matching records.
5. Count events and display destinations, ports, and timestamps.

CODE

```
logs = [
    ("10:00", "203.0.113.10", "192.168.1.5", 80),
    ("10:02", "198.51.100.8", "192.168.1.6", 443),
    ("10:05", "203.0.113.10", "192.168.1.7", 22)
]
threat_ips = {"203.0.113.10"}

print("THREAT INDICATOR MATCHES")
for t, src, dst, port in logs:
    if src in threat_ips:
        print(t, "| Source:", src, "| Destination:", dst, "| Port:", port)
```

OUTPUT



The screenshot shows a Visual Studio Code editor window with a file named `program_30.py`. The code defines a list of network logs and a set of threat IP addresses. It then iterates through the logs, checking if the source IP is in the threat set. If it is, it prints the timestamp, source, destination, and port. The terminal output shows the program running and displaying the matching records.

```
program_30.py
1 logs = [
2     ("10:00", "203.0.113.10", "192.168.1.5", 80),
3     ("10:02", "198.51.100.8", "192.168.1.6", 443),
4     ("10:05", "203.0.113.10", "192.168.1.7", 22)
5 ]
6 threat_ips = {"203.0.113.10"}
7
8 print("THREAT INDICATOR MATCHES")
9 for t, src, dst, port in logs:
10     if src in threat_ips:
11         print(t, "| Source:", src, "| Destination:", dst, "| Port:", port)
```

TERMINAL

```
PS C:\Forensics> python program_30.py
THREAT INDICATOR MATCHES
10:00 | Source: 203.0.113.10 | Destination: 192.168.1.5 | Port: 80
10:05 | Source: 203.0.113.10 | Destination: 192.168.1.7 | Port: 22
```

RESULT

The program compares network logs with threat indicators and displays matching connections.